STAT 410 / MATH 464 **Summer 2021**

Statistics and Probability II

Instructor: Alexey Stepanov E-mail: stepanov@illinois.edu

Office hours: Monday 9:00 – 10:00 a.m., Tuesday 8:00 – 9:00 p.m.,
Wednesday 11:00 a.m. – 12:00 noon, Thursday 6:00 – 7:00 p.m.,
Friday 2:00 – 3:00 p.m. (on Zoom) or by appointment.

Textbook: *Introduction to Mathematical Statistics,* Seventh Edition,
(not required) by Robert V. Hogg, Joseph W. McKean, Allen T. Craig.

Cumulative distribution function, moment-generating function, joint probability distributions, conditional distributions and expected values, functions of random variables, order statistics, gamma distribution, multivariate normal distribution, normal sampling theory, point estimation, maximum likelihood estimators, method of moments estimators, properties of estimators, convergence in probability, consistent estimators, convergence in distribution, central limit theorem, confidence intervals, sufficient statistics, efficiency of estimators, hypothesis testing, power of a hypothesis test, most powerful tests, Bayesian estimation.

The class lectures will be asynchronous (pre-recorded), the videos will be posted on Illinois Media Space: https://mediaspace.illinois.edu/channel/STAT 410 Summer 2021/ (with links on Compass2g). Some (most) examples will also be posted on Compass2g. Instructor and Teaching Assistant office hours will be held on Zoom: https://illinois.zoom.us/. We will also be using Piazza for class discussion: https://piazza.com/illinois/summer2021/stat410.

Examinations: There will be **three** exams worth 60 points each. The exam dates are:

Exam 1: Wednesday, July 7, 7:00 – 8:50 p.m. CDT

Exam 2: Wednesday, July 28, 7:00 – 8:50 p.m. CDT

Exam 3: TBA during Final Examination period (August 6–7).

Exams will be administered with online proctoring with Zoom and CBTF proctors (CBTF Online Student Guides: https://cbtf.engr.illinois.edu/cbtf-online/students/) and submitted on Gradescope: https://www.gradescope.com/.

Please login to https://cbtf.engr.illinois.edu/sched to automatically pick up your roster affiliation to your courses in the scheduler.

If you are unable to take an exam, you must contact the instructor in advance. All excuses must be verifiable. The make-up exams will be given only under exceptional circumstances.

Homework: There will be **eighteen** homework assignments worth 8 points each that will be given during the term as we cover the corresponding material. The lowest **three** homework scores will be dropped. Homework must be submitted by 5:00 p.m. CDT on the day it is due on Gradescope: https://www.gradescope.com/. **Late homework will NOT be accepted by the instructor or the TA regardless of the reason.**

Please put you final answers at the end of your work and mark them clearly. Show all work leading to your answers. No credit will be given without supporting work.

Homework assignments are meant to be learning experiences. You may discuss the exercises with other students in the class, but you must write-up the solutions on your own.

Policies:

- The exams are closed book and closed notes. For an exam, you are allowed to use **one** 8.5" by 11" sheet (both sides) with notes.
- You are allowed to use calculators for exams and homework assignments.
- Please put you final answers at the end of your work and mark them clearly.
- Show all work leading to your answers. **No credit will be given without supporting work.**
- The answers must be in a closed-form expression.
- Decimals answers must contain at least **four** significant digits.
- If the answer is a function, and its support must be included.
- You do not have to simplify the *function* answers after evaluating integrals, even though it is a polite thing to do.

Course Grade: The final course grade will be based on the total number of points earned by a student during the term in

Exams	–	$3 \times 60 = 180$ points
Homework	–	$15 \times 8 = 120$ points
Total	**–**	**300 points**

The grading scale will be as follows:

A+	TBD – 100% TBD – 300	A	93% – TBD 279 – TBD	A–	90% – 92.9% 270 – 278.5
B+	87% – 89.9% 261 – 269.5	B	83% – 86.9% 249 – 260.5	B–	80% – 82.9% 240 – 248
C+	77% – 79.9% 231 – 239.5	C	73% – 76.9% 219 – 230.5	C–	70% – 72.9% 210 – 218.5
D+	67% – 69.9% 201 – 209.5	D	63% – 66.9% 189 – 200.5	D–	60% – 62.9% 180 – 188.5

Grades are not curved or adjusted. This is not to dishearten students, but to let them know that their grade is based on individual effort and not on comparative effort.

Academic Integrity: The official University of Illinois policy related to academic integrity can be found in Article 1, Part 4 of the Student Code. Section 1-402 in particular outlines behavior which is considered an infraction of academic integrity: http://studentcode.illinois.edu/article1_part4_1-402.html.

In short, **do not cheat**. Given the situation, detecting cheating will be difficult. If we do detect cheating, that means it will have been an extremely flagrant violation, and we will seek the maximum possible punishment.

The instructor reserves the right to make any changes he considers academically advisable. Such changes, if any, will be announced through Compass2g. Please note that it is your responsibility to keep track of the proceedings.